MILITARY SPECIFICATION

CARTRIDGE, HE, 60MM, M49A4E1
LOADING, ASSEMBLING AND PACKING

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of high explosive mortar cartridge designated as Cartridge, HE, 60MM, M49A4E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

DRAWINGS

PICATINNY ARSENAL

PRODUCT AND PACKAGE DRAWINGS

9241292 - Cartridge, HE, 60MM, M49A4E1
9280108 - Box, Wirebound, Packing, Ammunition for 60MM Mortar Ammunition

FSC: 1310

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9280109 - Container, Ammunition, Metal: PA70 for 60MM Mortar Ammunition
9280110 - Container, Ammunition, Fiber, PA73 for 60MM Cartridge HE, M49A4E1 or Smoke, WP, M302A1E1

INSPECTION EQUIPMENT DRAWINGS

7256281 - Alignment Check Plug
7256348 - Alignment Gage
7257962 - Limit Check Gage
9221781 - Envelope Drawing for Leakage Testing

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with applicable drawings and specifications.

3.1.1 Clean explosive scrap.-Clean scrap (see 6.4) produced in regular operations may be remelted and reused provided that it complies with the requirements, except for form, for the grade of high explosive specified. Scrap from floor sweepings, ventilation, and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used.

3.2 Cartridge.-The cartridge shall comply with all requirements specified on drawing (dwg) 9241292, and with all requirements specified in applicable specifications.

3.3 Surface cavitation.-Cavities formed due to chipped or broken explosive from the sidewalls of the fuze well shall not exceed in aggregate more than 1/4 of the circumference. The maximum (max) dimension of any visible cavity in the faced (bottom) surface of the fuze well shall not exceed 1/4 inch in any direction and the total area of all cavities shall not exceed 80 percent of the faced surface area. No individual point of explosive shall extend above the flat surface on which the liner would rest. (Cavities having a max. dimension of 1/16 inch or less shall be disregarded in this computation.)

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3.4 Proving ground test.-The cartridges shall comply with the following requirements (see 4.2.3.2).

3.4.1 Misfires.-The cartridges shall not fail to fire from the weapon.

3.4.2 Duds.-The cartridges shall function on impact.

3.4.3 Short rounds.-No cartridge shall have a horizontal range less than 80 percent of the average horizontal range when calculated without the short rounds.

3.4.4 Permature burst.-There shall be no premature burst in the gun bore or in flight.

3.4.5 Metal parts separation.-There shall be no metal parts separation in the gun bore or in flight.

3.4.6 Range precision.-The horizontal range standard deviation shall be no greater than 7 meters when fired at charge 0 (Zero) and no greater than 30 meters when fired at charge two.

3.5 Workmanship.-Parts and assemblies shall be free of burrs, chips, dirt, grease, rust, corrosion and other foreign material.

3.6 Foreign matter in projectile cavity.-There shall be no foreign matter in the projectile cavity prior to high explosive pouring. Solid explosive shall not be present.

3.7 High explosive on threads.-Threads shall be free of high explosive.

3.8 High explosive on projectile body.-Exterior surface of the projectile body shall be free of high explosive.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Quality conformance inspection

4.2.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of cartridges shall contain:

a. Projectile metal part assemblies of one lot interfix from one manufacturer.

b. Composition B of one lot interfix from one manufacturer.

c. Fin assemblies of one lot interfix from one manufacturer.

d. Ignition cartridges of one lot from one manufacturer.

e. Propellant charges of one lot from one manfacturer.

f. Obturating rings of one lot interfix from one manufacturer.

g. Fuzes of one lot interfix from one manufacturer.

4.2.2 Examination.-See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48180 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9236378
4.2.2.1	Body Loading Assembly, Prior to Loading				NEXT HIGHER ASSEMBLY 9241292
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Foreign object in charge cavity		100%	3.6	Visual
Major A	None defined				
Major B	None defined				
Minor					
201.	Masking missing from obturating band area		0.65%	3.2	Visual
202.	Interior protective coating with bare spot		0.65%	3.2	Visual
203.	Evidence of poor workmanship		0.65%	3.2	Visual

NOTES:

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MIL-C-48180 (PA)

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48180 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9236378
4.2.2.2	Body Loading Assembly, After Loading				NEXT HIGHER ASSEMBLY 9241292
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	High explosive on projectile body		100%	3.8	Visual
Major A					
101.	Depth of fuze well, max		0.40%	3.2	Gage
102.	Diameter of fuze well, max		0.40%	3.2	Gage
Major B	None defined				
Minor					
201.	Surface cavitation excessive		0.65%	3.3	Visual
202.	High explosive on threads		0.65%	3.7	Visual
203.	Thread coating missing or inadequate		0.65%	3.2	Visual
204.	Evidence of poor workmanship		0.65%	3.5	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-48180 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.2.2.3	Cartridge				9241292
					NEXT HIGHER ASSEMBLY
					NA
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Ignition cartridge assembly above rear face of fin		100%	3.2	Gage
2.	Fuze pull wire missing or insecure		100%	3.2	Visual
3.	Propellant charges missing or damaged to the extent that propellant can escape		100%	3.2	Visual
4.	Obturating ring missing		100%	3.2	Visual
Major A					
101.	Obturating ring not flush or below bourrelet diameter		100%	3.2	Visual
102.	Outside diameter, max		100%	3.2	Gage
103.	Cartridge assembly fails to fully enter profile and alignment gage		100%	3.2	7256348/7256281/7257962
104.	Fin distorted		100%	3.2	Gage
105.	Obturating band damaged		0.40%	3.2	Visual
106.	Fin assembly torque		0.40%	3.2	Test
107.	Fuze torque		0.40%	3.2	Test
108.	Ignition cartridge torque		0.40%	3.2	Test
109.	Fin assembly not fully seated		0.40%	3.2	Visual
110.	Fuze not fully seated		0.40%	3.2	Visual
111.	Thread sealing compound does not cover a minimum of one half the number of male shell body threads for a full 360 degrees (prior to assembly)		0.40%	3.2	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-48180 (PA)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9241292
4.2.2.3	Cartridge				NEXT HIGHER ASSEMBLY NA
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
112.	Thread sealing compound does not cover a minimum of one half the number of male primer threads for a full 360 degrees (prior to assembly)		0.40%	3.2	Visual
113.	Thread sealing compound does not cover a minimum of one half the number of male fuze threads for a full 360 degrees (prior to assembly)		0.40%	3.2	Visual
Major B	None defined				
Minor					
201.	Paint damage area in excess of 1/4 square inch		0.65%	3.2	Visual
202.	Marking misleading or unidentifiable		0.65%	3.2	Visual
203.	Evidence of poor workmanship		0.65%	3.5	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-48180 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9280110
4.2.2.4	Fiber Container (Unsealed)				NEXT HIGHER ASSEMBLY 9280109
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101.	Nose support missing		0.40%	5.1	Visual
102.	Cushion(s) missing		0.40%	5.1	Visual
Major B	None defined				
Minor					
201.	Round cannot be removed by hand		0.65%	5.1	Manual
202.	Glue or asphalt on round		0.65%	5.1	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48180 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9280110
4.2.2.5	Fiber Container (Sealed)				NEXT HIGHER ASSEMBLY 9280109
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101.	Container damaged or cut through all asphalt impregnated layers		0.40%	5.1	Visual
102.	Sealing strip incomplete, badly wrinkled or missing.		0.40%	5.1	Visual
103.	Metal end loose or distorted		0.40%	5.1	Visual
Major B	None defined				
Minor					
201.	Marking missing, misleading or unidentifiable		0.65%	5.1	Visual
202.	Contents loose		0.65%	5.1	Manual

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CLASSIFICATION OF DEFECTS & TESTS MIL-C-48180 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9280109
4.2.2.6	Metal Ammunition Container (Unsealed)				NEXT HIGHER ASSEMBLY 9280108
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101.	Any material containing condensate and/or surface moisture		0.40%	5.2	Visual
102.	Spacer missing		0.40%	5.2	Visual
103.	Cushion missing		0.40%	5.2	Visual
104.	Incorrect number of assemblies		0.40%	5.2	Visual
Major B	None defined				
Minor	None defined				

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-48180 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9280109
4.2.2.7	Metal Ammunition Container (Sealed)				NEXT HIGHER ASSEMBLY 9280108
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B	None defined				
Minor					
201.	Marking missing, misleading or unidentifiable		0.65%	5.2	Visual
202.	Contents loose		0.65%	5.2	Manual

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CLASSIFICATION OF DEFECTS & TESTS MIL-C-48180 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9280108
4.2.2.8	Packing Box (Sealed)				NEXT HIGHER ASSEMBLY NA
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B	None defined				
Minor					
201.	Boards split or separated exposing containers		0.65%	5.2	Visual
202.	Contents loose		0.65%	5.2	Manual
203.	Marking misleading or unidentifiable		0.65%	5.2	Visual

NOTES:

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4.2.3 Testing

4.2.3.1 Air leakage test (see dwg. 9280109) Major A defect. The sealed metal ammunition container shall be tested 100 percent for leakage. Any container which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.3.1.

4.2.3.2 Proving ground test(see 3.4).-The cartridges shall be observed for the following characteristics:

a. Misfires, Major A defect (see 3.4.1).
b. Duds, Major A defect (see 3.4.2).
c. Short rounds, Critical defect (see 3.4.3).
d. Premature burst, Critical defect (see 3.4.4).
e. Metal parts separation, Critical defect (see 3.4.5).
f. Range precision, Major A defect (see 3.4.6).

Testing shall be as specified in 4.3.2.

4.2.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, eighty random sample cartridges shall be selected from each lot for this test. The lot shall be rejected if:

a. One or more misfires occur
b. Four or more duds occur
c. One or more short rounds occur
d. One or more premature burst occurs
e. One or more metal parts separation occurs
f. Horizontal range standard deviation at charge 0 or charge 2, multiplied by the factor 0.79 exceeds the applicable requirement.

4.2.3.2.2 After three consecutive lots have complied with the criteria of 4.2.3.2.1, forty-eight random sample cartridges shall be selected from each lot for this test. The lot shall be rejected if:

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a. One or more misfires occur
b. Four or more duds occur
c. One or more short rounds occur
d. One or more premature burst occurs
e. One or more metal parts separation occurs
f. Horizontal range standard deviation at charge 0 or charge 2, multiplied by the factor 0.74 exceeds the applicable requirement.

4.2.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.2.2.1 and 4.3 (see 6.2).

4.3 Test methods and procedures

4.3.1 Air leakage test.-The shipping and storage containers shall be tested in accordance with drawing 9221781.

4.3.2 Proving ground test.-The test shall be performed at a Government Proving Ground in accordance with the applicable USATECOM Acceptance Test Procedure.

The cartridges shall be fired from a standard 60mm M19 mortar at an elevation of forty-five degrees. One half of the sample cartridges shall be fired at charge zero and the other half at charge two. All fuzes shall be set for superquick functioning.

The individual velocity, pressure and range shall be recorded on the firing record.

Observation shall be made for misfires and duds, and shall be recorded. Should a premature burst or metal parts separation occur, a photograph shall be taken of the evidence and attached to the firing record. The velocity and range standard deviations and arithmetic averages shall be calculated and recorded. Range probable error shall be calculated and recorded. Temperature of the day, weather and the time of firing shall also be recorded.

Hawk radar shall be used to observe for metal parts separation, unobserved short rounds and unobserved duds. In these events, an analysis should be made of the radar data and should be noted on the firing record. This shall include a comparison of times of flight of the unobserved rounds with that of the normal rounds. Prints of the radar traces of these rounds, and

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prints of the preceding and succeeding rounds, shall be attached to the firing record. Data from ballistically acceptable lots may be discarded.

A smear camera shall be used to photograph all test and reference samples. The camera shall be positioned to pick up each round just outside the second velocity coil. Should an unobserved dud, short round or any other abnormal performance occur, a photo print of these rounds, and prints of all the other rounds in that particular charge zone, shall be attached to the firing record. Film from acceptable lots may be discarded.

4.3.2.1 Test validity.-If for any reason the proving ground or Picatinny Arsenal considers that the test conditions have detrimentally affected the test result, the test results shall be declared invalid and a new test shall be performed on additional sample cartridges.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-Packaging shall be as specified on Dwg. 9280110.

5.2 Packing and marking.-Packing and marking shall be as specified on Dwgs. 9280108 and 9280109.

6. NOTES

6.1 Ordering data.-See MIL-A-48078 (PA).

a. Ammunition data cards shall be prepared in accordance with MIL-STD-1167.

6.2 Submission of designs for approval.-See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 Submission of test data.-One copy of proving ground ballistic acceptance test data (i.e., firing record) shall be forwarded to each of the following:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-A, SARPA-AD-E-A4
Dover, New Jersey 07801

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6.4 Clean scrap.-Clean scrap is defined as scrap formed in kettles, pails, risers and loading machines used in the pouring and cooling operations, and scrap formed during the drilling operation. Scrap from floor sweepings, ventiliation, and suction apparatus (other than that of drilling operations) is not defined as being clean scrap and shall not be used.

6.5 Loading of high explosive.-Solid explosive shall not be added to the melted explosive or to the projectile body during pouring or cooling.

Custodian:
ARMY-PA

Preparing Activity:
ARMY-PA

Project Number: 1310-A024

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